

MAS Solution Strategies · Independent Review Record

Review method: Each set of notes is reviewed by an independent sub-agent. The reviewer did not take part in writing, works with fresh context, and only reads—does not edit. The basis for checking includes: the original "Specific behaviours" in the marking key for each question in [evidence/](#), the full 2016–2025 exam reports, the PDF pages of the original papers and keys, with independent calculations using sympy/scipy/numpy.

Handling principles: The author reviews each review comment one by one. All high-severity comments are checked back against the original paper text; anything that can be verified has been re-verified. Items marked "doubtful" by the reviewer are only changed when supporting evidence is found.

Conclusion categories: Accepted = changed as per the comment; Partially accepted = wording changed but the suggestion was not followed completely; Rejected = not changed, with reasons stated.

01 Complex numbers (12 items: accepted 12)

#	Severity	Issue	Action
1	High	Regular hexagon written incorrectly: in 2025F-Q7, z_0 is at the origin, the hexagon is not centred at O, and the product is $6r^5$, not r^5	Accepted: the author recalculated (moduli in order are $r, \sqrt{3}r, 2r, \sqrt{3}r, r$) and corrected it; the question-setting skill was also corrected
2	Medium-high	2025A-Q19(b) omits the periodic solutions $n = 3, 4, 5 + 12k$	Accepted
3	Medium	Maximum argument: $\sin \theta = r/d$ does not give the argument	Accepted
4	Medium	2020A-Q10(b): the ray endpoint i is on the locus (solid point)	Accepted
5	Low	The closest-distance formula only holds when the point is outside the circle	Accepted
6	Low	"Always use polar form for multiplication and division" contradicts "use the conjugate for division"	Accepted
7	Low	Report year and content mismatch ($-z$ is from the 2021 report)	Accepted
8	Low	The third-quadrant example is not the original report wording; $5\pi/4$ cannot be written either	Accepted
9	Low	Missing citation of 2022F-Q6; the sample paper is not specified	Accepted
10	Low	The "CF opening / CF hard question" labels do not match the cited CA questions	Accepted
11	Low	" i^n is periodic with period 4" is inaccurate ($n \cdot i^n$ is not periodic)	Accepted
12	Low	When $\theta < 0$, it should be written as clockwise by $ \theta $	Accepted

02 Functions and graphs (16 items: accepted 16)

#	Severity	Issue	Action
1	High	Reciprocal-function table " $f \rightarrow \pm\infty \Rightarrow$ horizontal asymptote $y = 0$ " is wrong: a vertical asymptote of f corresponds to an open point $(b, 0)$ of $1/f$	Accepted: split into two rows
2	Medium	The table omits "horizontal asymptote $y = c \rightarrow y = 1/c$ "	Accepted
3	Medium	"Hole-to-hole" only holds when $b \neq 0$	Accepted
4	Medium	The original 2016A-Q12(b) is $ f(x) = d$, not $f(x) = d$	Accepted

5	Medium	"Flat bottom of two V-shaped lines" is described incorrectly	Accepted: changed to the flat bottom of the sum of two absolute values; the author calculated $a = 3$, $k = 2$
6	Low	The integral of the W-shaped graph is one triangle, not two	Accepted
7	Medium	The sign of the inverse function should be determined by $\text{ran } f^{-1} = \text{dom } f$	Accepted
8	Medium	When a square root is in the denominator, it must be > 0 (2025F-Q2(a))	Accepted
9	Medium	" $c > 0$ guarantees no zeros" is not enough; a and c must have the same sign	Accepted
10	Low	"Turning point (if the question requires it)": 2017F-Q5 awarded marks even though it was not required	Accepted
11	Low (doubtful)	The fabricated " $c = 2$ touches the x-axis" example conflicts with the meaning of c in the actual question	Accepted: used the original question wording instead
12	Low	"Most stable in the whole course" is an overgeneralisation (in 2018 it was instead listed as well answered)	Accepted
13	Low	"Do not use it" omits the 2019 report	Accepted
14	Low	The example $\ln x - 3 $ does not come from the 2022 report	Accepted
15	Low	Missing citations of 2016F-Q8(b), 2022F-Q2(a)	Accepted
16	Low	"Intersection points unchanged" is vague	Accepted

03 3D vectors and systems of linear equations (13 items: accepted 13)

#	Severity	Issue	Action
1	High	Laser reflection direction written incorrectly: the vector obtained by adding in the $S \rightarrow B$ direction is parallel to the mirror; it should be the sum of the unit vectors of $B \rightarrow S$ and $B \rightarrow R$, or $\vec{d}_{\text{out}} - \vec{d}_{\text{in}} \parallel \vec{n}$	Accepted: the author used numpy to verify $\vec{d}_{\text{out}} - \vec{d}_{\text{in}} = (0, 0.23, 0.46) \parallel (0, 1, 2)$
2	High	2024A-Q16(c) is the shortest distance between two lines (two independent parameters), while 2017A-Q14 is solving an inequality to find the duration; the two questions were conflated	Accepted: split into two categories
3	Medium	"CF is tested every year" is untrue (not tested in 2017 or 2019; in 2021 it was in CA)	Accepted
4	Medium	"Named for five consecutive years" does not match the reports; 2024 is missing	Accepted
5	Medium	"Must write parametric form" is an overgeneralisation; the key accepts representation using variable relationships	Accepted
6	Medium	"Wavy lines, dot product, cross product" is not the original report wording	Accepted: marked as author's suggestion
7	Low-medium	The point-to-plane distance formula is not on the formula sheet	Accepted
8	Low	"Many people have never seen it" is an exaggerated paraphrase	Accepted
9	Low	The line-plane angle formula lacks absolute value, making it inconsistent	Accepted
10	Low	The geometric interpretation table omits "three planes coincident"; 2021A-Q14 requires both points to be written	Accepted
11	Low	The letters a and d are each used for two things	Accepted
12	Low	"Final CA question" does not match 2018F-Q8	Accepted
13	Low (doubtful)	Sample paper question not specified	Accepted

04 Vector calculus and motion (12 items: accepted 11, partially accepted 1)

#	Severity	Issue	Action
1	High	2019A-Q13(d) is not a parabola; it is a figure-eight shape	Accepted
2	High	The three examples of value restrictions all disagree with the key (2024A-Q18 requires $x \geq 2$ and $y \geq 0$)	Accepted: replaced with the original key wording
3	Medium	"One loop = 0 to 2π " is an overgeneralisation (2017A-Q15 is 6π , 2019A-Q13 is 4π)	Accepted
4	Medium	2021A-Q19 has no wind and no launch angle, but has lift	Accepted: split into two models
5	Medium	"Uniform velocity" should be "constant speed"	Accepted
6	Medium	The descent angle $90^\circ - \phi$ gives a negative value ($\phi = 143.3^\circ$)	Accepted: changed to $ 90^\circ - \phi $; the author verified 53.3°
7	Low	"Three scoring points" is an overgeneralisation	Accepted
8	Low	The height is relative to the centre; 100 m above the ground corresponds to $y = 20$	Accepted
9	Low	The order of subtraction is not given a source or rule	Accepted
10	Low	In 2024A-Q16(c), both speed and departure time are adjustable	Accepted
11	Low	Parameter elimination omits the "x is linear in t" type (named in the 2018 report)	Accepted
12	Low (doubtful)	The SHM in 2019A-Q18(d) is not mentioned	Partially accepted: added a cross-reference; the method is in the differential equations notes

05 Integration techniques and applications (11 items: accepted 11)

#	Severity	Issue	Action
1	Medium	"Irreducible quadratic" was paired with the reducible $x^2 + 3x + 1$ (discriminant $5 > 0$)	Accepted: replaced with $x^2 + 2$; a separate note was added for 2022F-Q4
2	Medium	The error example from the 2021 report is paraphrased incorrectly (the original is $\sqrt{2 - \sin^2\theta} \neq \sqrt{2} - \sin\theta$)	Accepted
3	Medium	"Substitution is always tested in CF" is untrue (not tested in 2022), and there are 7-mark questions	Accepted
4	Medium	"Partial fractions are always tested every year" is untrue (not tested in 2017 or 2023)	Accepted
5	Medium	2023A-Q17(c) is an increment formula question; 2022A-Q17 does not use dV/dh	Accepted: split and rewritten
6	Medium	"Absolute value is a separate 1 mark" is untrue; the absolute value is included in the antiderivative mark	Accepted
7	Low	The "correct notation" mark is awarded together with the limits; the area expression does not need π	Accepted
8	Low	2016A-Q9 is an indefinite integral; for 2021F-Q3 the substitution must be chosen yourself	Accepted
9	Low	3 questions are not categorised	Accepted
10	Low	The 4 in the cardioid is the product of two parts	Accepted
11	Low	The 2024 report says the numerator after finding a common denominator was written incorrectly, not that comparing coefficients was wrong	Accepted

06 Rates of Change and Differential Equations (15 items: 15 accepted)

#	Severity	Issue	Action
1	Medium	The main solution in the key for 2017A-Q17(b) is to count periods, not $\int v $	Accepted
2	High	The point of inflection in 2025A-Q16 is at the maximum slope (3, -1), not where the slope is 0	Accepted
3	Medium	'10% growth per year' should be an instantaneous growth rate ($dP/dt = 0.1P$); otherwise the answer changes from 333 to about 412	Accepted: editor verified both results
4	Low	For 2025A-Q15, $x(0)$ is -1	Accepted
5	Medium	The letter k refers to carrying capacity and the constant of proportionality in different questions, so it is easily confused	Accepted
6	Low	'Initial value above $k/2$ is concave down throughout' needs $P_0 < k$ added	Accepted
7	Low (doubtful)	The SHM criterion should allow for a shifted equilibrium point $a = -n^2(x - c)$	Accepted
8	Low	The wording 'cannot differentiate directly with respect to t' is unclear	Accepted
9	Low	2024A-Q12 does not give a direction; both $\pm A \sin$ are awarded marks	Accepted
10	Medium	The marking panel reminder missed 2024 (logistic), 2017 (standard equation), and 2022 (absolute value)	Accepted
11	Low	The exact phrase 'integration statement' appears in only two years	Accepted
12	Low	2016A-Q18 is $a - bx$; the two conditions come from different sources	Accepted
13	Low	'keep only one independent variable' contradicts the shadow question; the past-paper list has incorrect inclusions and omissions	Accepted
14	Low	A vertical tangent only holds when the numerator $\neq 0$	Accepted
15	Low	2018A-Q20 says 'do not' solve, not 'need not'	Accepted

07 Statistical Inference (16 items: 16 accepted)

#	Severity	Issue	Action
1	High	2022A-Q12(d) is 'the same sample duplicated', not two samples combined; this conflicts with the conclusion of 2019A-Q15(c)	Accepted: write the two cases separately
2	Medium (doubtful)	'the key accepts both' is unsupported; the key only gives $X < T/n$	Accepted; question-setting skill corrected accordingly
3	Medium	'examined every year' is untrue (not examined in 2016, 2020)	Accepted
4	Medium	Missing counterexample 2025A-Q13(e) (many intervals not containing μ is the evidence)	Accepted
5	Low	'citing the CLT' is not a standalone marking condition in the key	Accepted
6	Low	The calculation using 304.4 and 2.326 is inconsistent (should be 304.3)	Accepted: editor recalculated
7	Low	Rounding when back-solving for n: the key accepts both 2.576 and 2.58; add 2024A-Q14(f)	Accepted
8	Low	Missing the one-tailed and half-interval cases	Accepted
9	Low	2024A-Q14(c) does not ask for n	Accepted
10	Low	'the answer is unchanged' applies only to 2018A-Q12(c)	Accepted
11	Low	For the graphing question, the mark for labelling the probability region was omitted	Accepted
12	Low	The key's reason for 2024A-Q14(e) is that the sample sizes are different	Accepted
13	Low	Sample paper question not explained; sample paper A9 is also not referenced	Accepted
14	Low	'definitely' should be 'very likely'	Accepted

15	Low	The precision table lacks context ($\mu = 80$); s and σ are used interchangeably	Accepted
16	Low	' $n \geq 30$ applies' is an overgeneralisation; when the population is normal, any n works	Accepted

Summary (7 MAS documents): 95 comments in total (including 5 'doubtful' items), 94 accepted, 1 partially accepted, and 0 rejected. There were 8 high-severity items; all were revised after returning to the original paper text or recalculating.

Classification corrections after user feedback (2026-10-09)

- User feedback: the 'Vectors' section contains unrelated differential equations questions. This was verified: 2017A-Q18 (carousel, cosine rule + implicit differentiation) and 2019A-Q18 (Ferris wheel, related rates + SHM proof) contain no vector content and were originally misclassified under vector calculus as 'circular motion'. They have been moved out and kept only in 'Rates of Change and Differential Equations'; the corresponding section in the notes has also been moved across. Vector calculus 14 $\rightarrow$ 12 questions.
- 2021A-Q19 (ski jump) gives the position vector $r(t)$ and asks for the Cartesian path and landing angle; it is vector motion, so it is retained.
- A keyword review was carried out on all tags; MAS found no other misclassifications.